

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

303000088 - Aprendizaje Automático: Modelos Y Aplicaciones

DEGREE PROGRAMME

30AH - Máster Universitario En Matemáticas Avanzadas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

Index

Learning guide

1. Description.....	1
2. Faculty.....	1
3. Prior knowledge recommended to take the subject.....	2
4. Skills and learning outcomes	2
5. Brief description of the subject and syllabus.....	3
6. Schedule.....	6
7. Activities and assessment criteria.....	9
8. Teaching resources.....	10
9. Other information.....	11

1. Description

1.1. Subject details

Name of the subject	303000088 - Aprendizaje Automático: Modelos y Aplicaciones
No of credits	6 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	30AH - Máster Universitario en Matemáticas Avanzadas
Centre	30 - Esc. Politéc. Enseñanza Superior (epes)
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Oscar Araque Iborra	ETSIT, C-212	o.araque@upm.es	Sin horario. Acordar la tutoría por correo electrónico
Pedro Jose Zufiria Zatarain (Subject coordinator)	ETSIT, A-306	pedro.zufiria@upm.es	Sin horario. A acordar con el profesor mediante correo electrónico.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Probability Theory fundamentals
- Multivariable Analysis fundamentals
- Statistical Inference fundamentals
- Linear Algebra fundamentals

4. Skills and learning outcomes *

4.1. Skills to be learned

CO3 - Conocer diversas aplicaciones de las teorías matemáticas y su modelización en otros entornos o disciplinas que permitan adaptar los conocimientos matemáticos a nuevos problemas. TIPO: Conocimientos o contenidos.

CP1 - Utilizar programas informáticos especializados como laboratorio para diseñar, construir o validar hipótesis científicas en alguno de los ámbitos de las matemáticas. TIPO: Competencias.

CP2 - Ser capaz de trabajar en equipos multidisciplinares que involucren conocimientos matemáticos avanzados en áreas teóricas o aplicadas. TIPO: Competencias.

CP4 - Comunicar de forma oral y escrita conocimientos y conclusiones científicas, valorando diferentes interpretaciones en públicos especializados y no especializados de una manera clara y rigurosa. TIPO: Competencias

CP6 - Reflexionar sobre la relevancia social y ética de la aplicación de los conocimientos adquiridos a determinados problemas. TIPO: Competencias

HA1 - Aplicar los conocimientos matemáticos adquiridos a la resolución de problemas nuevos de una manera crítica, generando hipótesis de trabajo claras, definiciones precisas, nuevos enfoques con fundamento matemático y soluciones originales. TIPO: Habilidades o destrezas

HA4 - Poseer destreza para proponer, validar e interpretar modelos matemáticos de diferente dificultad en situaciones reales complejas, utilizando las herramientas matemáticas más adecuadas a los fines que se persigan, y valorando las hipótesis y conclusiones asociadas a cada nivel de dificultad del modelo. TIPO: Habilidades o destrezas

4.2. Learning outcomes

RA1 - To understand and apply the fundamental models in unsupervised, supervised, and reinforcement learning problems.

RA2 - Los resultados de aprendizaje en las tres categorías propuestas: conocimientos/contenidos, habilidades y competencias son los señalados en la memoria del título

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Machine learning is a discipline that provides a body of knowledge and associated modeling techniques to solve scientific and engineering problems (aimed to describe, predict or explain), in which we have data or experimental knowledge as a main resource.

This course covers the characterization of the basic types of learning problems and the study of the fundamental modelling paradigms aimed to solve them. To strengthen the understanding of the field of machine learning, the course follows a practical approach where the application of fundamental concepts and techniques is central to the development of the syllabus.

Students taking this course will learn to implement models and their respective evaluation in real-case scenarios using Python and its ecosystem of libraries and tools.

5.2. Syllabus

1. Introduction and Mathematical Foundations

- 1.1. Definition of problems. Model construction and inductive biases.
- 1.2. The machine learning problem. The case of statistical learning.
- 1.3. Mathematical foundations.
 - 1.3.1. Dynamical systems and iterative methods for optimization.
 - 1.3.2. Review of fundamental concepts in classical statistical inference.

2. Types of Statistical Learning Problems. Associated models

- 2.1. Types of statistical learning problems.
 - 2.1.1. Supervised learning: regression problems and classification problems.
 - 2.1.2. Unsupervised learning: the problem of density estimation and procedures aimed at its approximation.
 - 2.1.3. Reinforcement learning: the optimal control problem in partially known environments.
 - 2.1.4. Hybrid learning scenarios: semi-supervised learning.
- 2.2. A brief overview of standard associated modelling paradigms.

3. Models for Supervised Learning

- 3.1. Regression.
 - 3.1.1. Fundamentals of linear and nonlinear regression models.
 - 3.1.2. Artificial neural networks: Multilayer Perceptron and deep architectures.
 - 3.1.3. Other models for regression: K-Nearest Neighbors (K-NN) and models based on Gaussian Processes.
- 3.2. Classification.
 - 3.2.1. Artificial neural networks revisited.
 - 3.2.2. Support Vector Machines.
 - 3.2.3. Decision trees and random forests.
- 3.3. Ensemble models.
- 3.4. Dynamical systems modelling. Recurrent architectures.

4. Models for Unsupervised Learning.

4.1. Dimensionality reduction techniques: Principal Component Analysis (PCA) and Singular Value Decomposition (SVD).

4.2. Clustering techniques: K-means, Hierarchical Clustering, DBSCAN and Self-Organizing Maps (SOM).

5. Practical Applications

5.1. Science and engineering: signal processing, black-box function optimization, time series forecasting, fault diagnosis, control of dynamic systems, and complex network analysis.

5.2. Computational social sciences: natural language processing and language modelling (e.g., recurrent networks, LLMs).

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	1.1 and 1.2. Introduction Duration: 02:00 Lecture 1.3. Mathematical foundations. Duration: 02:00 Lecture			
2	2.1 and 2.2. Supervised and unsupervised problems. Duration: 02:00 Lecture 2.3, 2.4 and 2.5. Reinforcement and hybrid learning. Modelling paradigms Duration: 02:00 Lecture			
3	3.1.1. Linear and nonlinear regression. Duration: 04:00 Lecture			
4	3.1.2. Artificial neural networks for regression. Duration: 02:00 Lecture Lab 1. Data processing Duration: 02:00 Laboratory assignments			
5	3.1.3. Other models of regression. Duration: 02:00 Lecture 3.2.1. Artificial neural networks for classification. Duration: 02:00 Lecture			
6	3.2.2. and 3.2.3. Support vector machines Duration: 02:00 Lecture Lab 2. Machine learning for regression Duration: 02:00 Laboratory assignments			

7	3.2.3. and 3.2.4. Decision trees and random forests. Recurrent architectures Duration: 04:00 Lecture			
8	4.1. and 4.2. Dimensionality reduction: PCA and SVD. Duration: 02:00 Lecture Lab 3. Machine learning for classification. Evaluation: AUC, etc. Duration: 02:00 Laboratory assignments			
9	4.2. Clustering Duration: 04:00 Lecture			
10	Lab 4. Artificial Neural Networks Duration: 02:00 Laboratory assignments			Evaluation of first part. Written test Progressive assessment Presential Duration: 02:00
11	3.3. Ensemble models. Duration: 04:00 Lecture			
12	3.2. and 3.4. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). Duration: 04:00 Laboratory assignments			
13	5. Applications of ANNs. Duration: 02:00 Lecture Lab 5. Application of unsupervised models. Duration: 02:00 Laboratory assignments			
14	5. Applications: Word embeddings, Language Modeling and LLMs. Duration: 04:00 Lecture			
15	5. Challenge development. Duration: 02:00 Laboratory assignments 5. Challenge presentation. Duration: 02:00 Laboratory assignments			
16				
				Evaluation of second part. Exam. Written test Progressive assessment Presential Duration: 02:00 Evaluation of second part. Challenge. Written test Progressive assessment Presential Duration: 01:00

17				Evaluation of practices. Problem-solving test Progressive assessment Presential Duration: 01:00 Global evaluation (first part) Written test Global examination Presential Duration: 01:00 Global evaluation (second part: Exam+Challenge) Written test Global examination Presential Duration: 03:00
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Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
10	Evaluation of first part.	Written test	Face-to-face	02:00	45%	3 / 10	HA4 CP6 HA1 CP2 CP4 CO3
17	Evaluation of second part. Exam.	Written test	Face-to-face	02:00	30%	3 / 10	
17	Evaluation of second part. Challenge.	Written test	Face-to-face	01:00	15%	3 / 10	CP1 CO3 HA4 CP6 HA1 CP2 CP4
17	Evaluation of practices.	Problem-solving test	Face-to-face	01:00	10%	3 / 10	CP1 CO3 HA1 CP2 CP4

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Global evaluation (first part)	Written test	Face-to-face	01:00	45%	5 / 10	
17	Global evaluation (second part: Exam+Challenge)	Written test	Face-to-face	03:00	45%	5 / 10	CP1 CO3 HA4 CP6 HA1 CP2 CP4

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Extraordinary Evaluation	Written test	Face-to-face	03:00	100%	5 / 10	

7.2. Assessment criteria

The evaluation will assess that the students have learnt the fundamental concepts of the course and have acquired the basic skills to address practical problems.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
K. Murphy. Machine learning: A probabilistic perspective. 2012	Bibliography	It addresses machine learning based on probabilistic and inference models.
A. Géron. Hands-On Machine Learning with Scikit-Learn, Keras & Tensor Flow	Bibliography	Introduces the basics of using machine learning techniques in Python with Scikit-Learn and other tools.
W. Goodfellow, J. Bengio and A. Courville, Deep Learning, 2016, HTML version available in: http://www.deeplearningbook.org/ .	Bibliography	The basic reference for Deep Learning models.

G. James, D. Witten, T. Hastie, R. Tibshirani, An Introduction to Statistical Learning, with applications in R, Springer Verlag, Berlin Heidelberg, 2014.	Bibliography	It presents some fundamentals of statistical learning, and illustrates them with applications developed in R.
C. Bishop, Pattern Recognition and Machine Learning, Springer 2006.	Bibliography	It presents with rigor and extension the mathematical foundations (especially the probabilistic ones) of different models to address machine learning problems.
S. Haykin, Neural Networks and Learning Machines, Prentice Hall, 2008.	Bibliography	It presents the fundamentals of machine learning using neural network models.

9. Other information

9.1. Other information about the subject

The course will be taught in English.

Besides the use of the Moodle platform, in case of need, the communication between professors and students will be carried out using Zoom.

The answers of the tests questions will not be published since each student answers different ones, randomly drawn from a pool of questions.

In this course, collaboration is encouraged, but in no way copying. Copying laboratories is considered academic fraud, which represents a serious infraction contemplated in the Learning Assessment Regulations of the Universidad Politécnica de Madrid. Also, the use of AI tools is prohibited and may lead to a disciplinary action. In that sense, any evaluation or submission is open to require a supplementary oral assessment by the teacher to validate that it has been made by the student without the help of AI systems.

Relationship with Sustainable Development Goals:

On the one hand, the course contributes to SDG 4 (sub-objective 4.4): Increase the number of young and adult people having professional and technical competences necessary to have access to employment and entrepreneurship.

On the other hand, the course studies fundamental mathematical and statistical tools to be employed in the modelling of different types of systems such as biological ones (SDG 3), data processing ones for industrial innovation and sustainable cities/communities (SDGs 9 and 11), climate ones (SDG 13) or ecosystems (SDGs 14 and 15).

In the course, collaboration is encouraged, but in no way copying. Copying laboratories is considered academic fraud, which represents a serious infraction contemplated in the Learning Assessment Regulations of the Universidad Politécnica de Madrid.